

A Data-Driven and Statistical Analysis of Off Campus College Transport, Inc. Ridership Trends, Campus Shuttle Directionality, and Main Street–Westside Corridor Demand

McKenzie Skrastins

Advisor: Dr. Xingye Qiao

IMP Capstone Project: Data Science Major

May 2026

Abstract

This project analyzes Off Campus College Transport, Inc. (OCCT) ridership data to identify patterns in student transportation behavior and evaluate possible service improvements. Using Tableau visualizations, inferred travel patterns, and negative binomial regression models, the report examines system-wide ridership trends, Campus Shuttle efficiency, reverse-direction shuttle feasibility, and the relationship between the Main Street and Westside corridors.

The analysis shows that OCCT ridership is highly concentrated around major routes, central campus stops, and weekday daytime/evening travel periods. University Union functions as the dominant system hub, while stops such as Mohawk, Engineering Building, Hillside, UCLUB, and University Downtown Center also show substantial activity. The reverse Campus Shuttle analysis suggests that reverse-direction demand exists for certain stop pairs, but not evenly across the entire route. The Main Street and Westside corridor analysis finds that Westside fullness is positively associated with expected Main Street boardings in both directions, suggesting that Main Street may function as a complementary corridor during periods of higher Westside demand. Overall, the findings suggest that future OCCT planning should consider route direction, stop-level demand, time of day, bus fullness, and corridor interactions rather than relying only on aggregate ridership totals.

Introduction

Public transportation plays an important role in how students move throughout and beyond a college campus. At Binghamton University, Off Campus College Transport, Inc, commonly known as OCCT, provides students with access to campus buildings, residential areas, commercial corridors, and off-campus neighborhoods. Because thousands of students rely on OCCT throughout the academic year, understanding ridership patterns is important for evaluating how well the system serves student needs and where future improvements may be possible.

This project analyzes OCCT ridership data to identify patterns in student transportation behavior across routes, stops, times of day, seasons, and directions of travel. Rather than beginning with a single hypothesis, this project takes an exploratory data analysis approach. The goal is to use historical ridership data to better understand how students use the bus system and to generate practical insights that may support future route planning, service adjustments, and resource allocation.

Overall, this report uses data visualization, descriptive analysis, and statistical modeling to summarize key findings from OCCT ridership data. The purpose is not only to describe past usage, but also to highlight patterns that may inform future decisions about shuttle design, corridor service, and student transportation needs.

Data and Methods

The following analysis focuses on three main areas. First, Tableau visualizations are used to examine broad trends in ridership, including stop activity, route frequency, hourly demand, and geographic patterns across the OCCT system. These visualizations help identify which locations experience the highest boarding activity and how demand varies over time. Tableau was used to create interactive graphics showing route frequency, stop activity, hourly demand, seasonal variation, and geographic ridership patterns. These graphics provide a broad overview of OCCT usage and help identify the routes, stops, and time periods with the highest levels of activity.

Second, the project investigates Campus Shuttle usage, with particular attention to inferred travel patterns and whether bidirectional service may be useful. Finally, the analysis compares the Main Street and Westside corridors to examine whether ridership on one corridor changes when the other corridor is more or less busy. These investigations were performed in R.

To understand this analysis, it is important to understand how the data was collected. When a rider enters the bus, they scan their Binghamton University ID and a record is logged internally within the university. These records are compiled into a dataset; the following analysis uses records from Fall 2023 to Winter 2025. Table ?? displays a list of variables that were recorded and used in this analysis:

Route Variables	Rider Variables
CorridorName - Name of the Route the rider was taking	PERSON_UID - Unique ID of the rider
StopName - Name of the stop the that rider entered from when they swipped their ID	Date - Day, month, and year which the rider entered the bus
VehicleID - Unique ID of the bus which the rider entered	Time - Hour and minute which the rider entered the bus
	Year - The year the rider entered the bus
	College - BU College of the rider
	Major - Major of the rider
	Program Level - Undergraduate or Graduate

Table 1: Key route and rider variables used in the OCCT ridership analysis.

Before the analysis was performed, the data were cleaned and transformed to make them more useful for visualization and modeling. Stop names were standardized where necessary, and stops with missing or unknown information were removed from parts of the analysis. Time variables were also extracted from the original timestamp so that ridership could be compared by hour of the day, minute of the hour, day of the week, month, and season. Stop locations were matched with latitude and longitude coordinates so that ridership patterns could be displayed spatially in Tableau. While the original data records when and where a rider boards the bus, it does not record where that rider exits. To study possible travel patterns, exit stops were inferred using consecutive boardings from the same rider when available. For example, if a student boarded at one stop and later, within the same day, boarded again from a different location, the second boarding location can provide an approximate indication of where the first trip may have ended. This approach does not perfectly capture every trip, but provides a useful approximation for identifying common travel flows and possible reverse-direction demand. These assumptions mean that the findings should not be treated as a perfect representation of all rider behavior. However, they still provide meaningful evidence about how students interact with the transit system and where further investigation may be valuable. The following variables were inferred using the raw, recorded variables. See table ?? ;

Inferred Route Variables	Inferred Rider Variables
Stop_base - Standardized version of StopName used to group similar stop labels that refer to the same general location, inferred using StopName	Season - Academic Semester type of when the rider entered the bus (fall, winter, spring, summer), inferred using Date
lat - Latitudinal coordinates of a stop, inferred using Stop_base	Day - The day the rider entered the bus, inferred using Date
lon - Longitudinal coordinates of a stop, inferred using Stop_base	Month - The month the rider entered the bus, inferred using Date
exit_stop - Stop at which the rider exited, inferred using Stop_base	Hour - The hour the rider entered the bus, inferred using Time
Route Type - The general Corridor the rider is taking (excluding Inbound and Outbound specificities), inferred from CorridorName	Minute - The minute the rider entered the bus, inferred using Time
Direction - whether the stop is Inbound or Outbound, inferred from CorridorName	DayOfWeek - The day of the week the rider entered the bus, inferred using Date

Table 2: Key inferred route and rider variables used in the OCCT ridership analysis.

These inferred variables make it possible to study both system-wide ridership patterns and more specific questions about particular shuttle routes and corridors.

The goal of this analysis was to better understand how riders use the OCCT system, where demand is concentrated, and whether the data suggests possible opportunities for service improvements. While the findings are limited by the available data and the assumptions required to infer exits, the analysis still offers useful insight into student transportation behavior and can serve as a foundation for future OCCT planning.

Exploratory Analysis with Visualizations

To begin the analysis, a series of Tableau dashboards were created to summarize broad patterns in the 2023–2025 OCCT dataset. These visualizations provide an overview of route usage, directional ridership, stop activity, and vehicle frequency before moving into more specific analyses of the Campus Shuttle and Main Street/Westside corridors. These graphics serve as a starting point for understanding the structure of the dataset and for identifying patterns that require deeper analysis.

Figure 1 summarizes general route demographics in the data. The route frequency chart shows that Campus Shuttle appears most often in the dataset, followed by UCLUB, Westside Outbound, Westside Inbound, and Downtown Center routes. This suggests that a

large share of recorded OCCT activity is concentrated among a relatively small number of high-use routes. The direction chart shows that inbound and outbound trips are fairly balanced overall, although inbound routes appear slightly more frequently than outbound routes. The stop frequency chart shows that University Union is by far the most common stop in the dataset, followed by Mohawk, UCLUB, Engineering Building, and Hillside. This reinforces the role of University Union as a major central hub in the OCCT system, while also suggesting that its unusually high level of activity may strongly influence analyses of stop-level ridership patterns. Finally, the vehicle frequency chart shows that some buses appear in the dataset much more often than others. Together, these visualizations show that OCCT ridership is not evenly distributed across routes, stops, or vehicles; instead, activity is heavily concentrated around major routes and central campus stops.

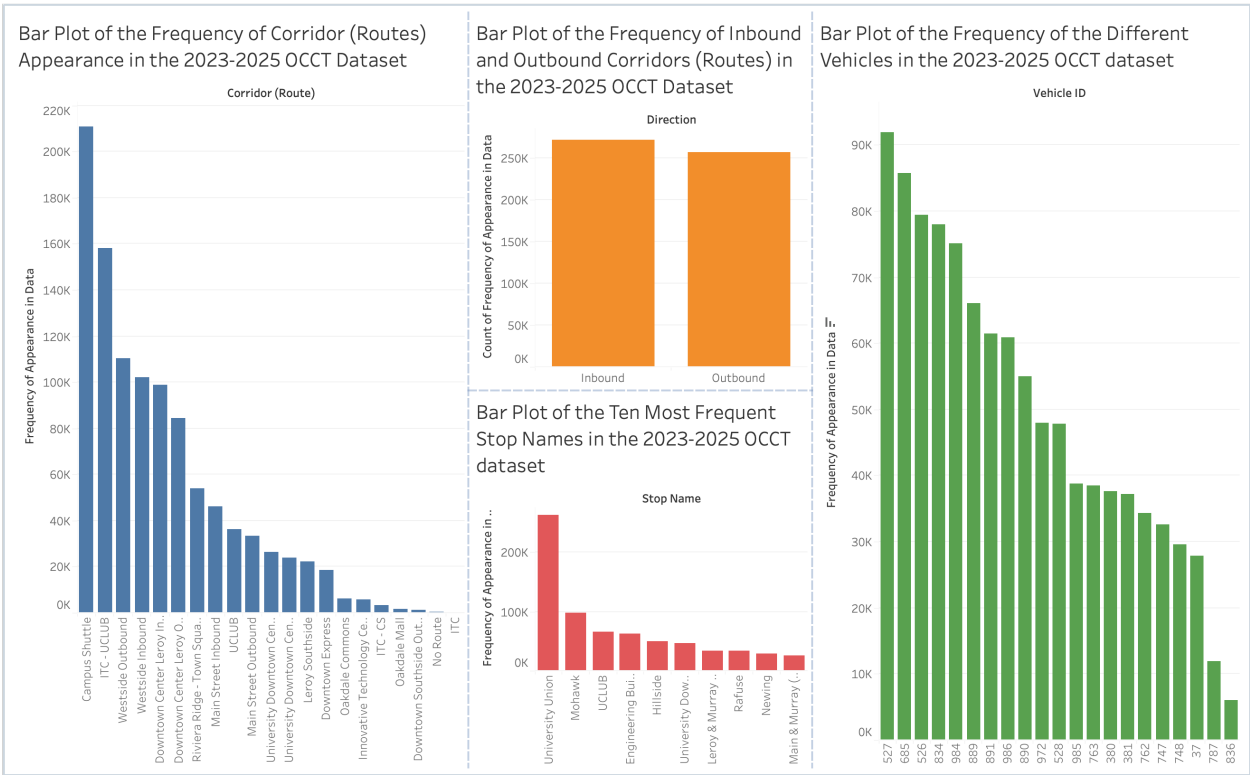


Figure 1: Dashboard of Route Demographics in the 2023-2025 OCCT Dataset - Route Demographics Interactive Dashboard

Figure 2 summarizes rider demographics by academic major, age, college, and program level. The major frequency chart shows that Computer Science riders appear most often in the dataset by a large margin, followed by Psychology, Biological Sciences, and Integrative Neuroscience. Although Computer Science is housed in Watson College, the college breakdown shows that undergraduate Harpur students make up the largest share of OCCT riders overall. This suggests that Harpur's larger student population and variety of majors may outweigh the high frequency of any single Watson major. The program-level pie chart also shows that undergraduate students make up the majority of recorded bus entries, while graduate students represent a smaller portion of the dataset.

Although one might expect OCCT ridership to be driven primarily by students traveling to satellite campuses, such as the Nursing school, the Pharmacy school, or the College of Community and Public Affairs (CCPA) which is housed in the Downtown Center, these visualizations suggest a different pattern. The age distribution is concentrated most heavily among 19–21-year-old riders, and the college breakdown is dominated by undergraduate Harpur and Watson students rather than satellite-campus programs. When combined with the previous dashboard, where University Union, UCLUB/ITC, and Westside inbound/outbound routes were among the most frequent stops and corridors, the data suggests that OCCT is used largely by traditional undergraduate students traveling between residential areas and the main campus.

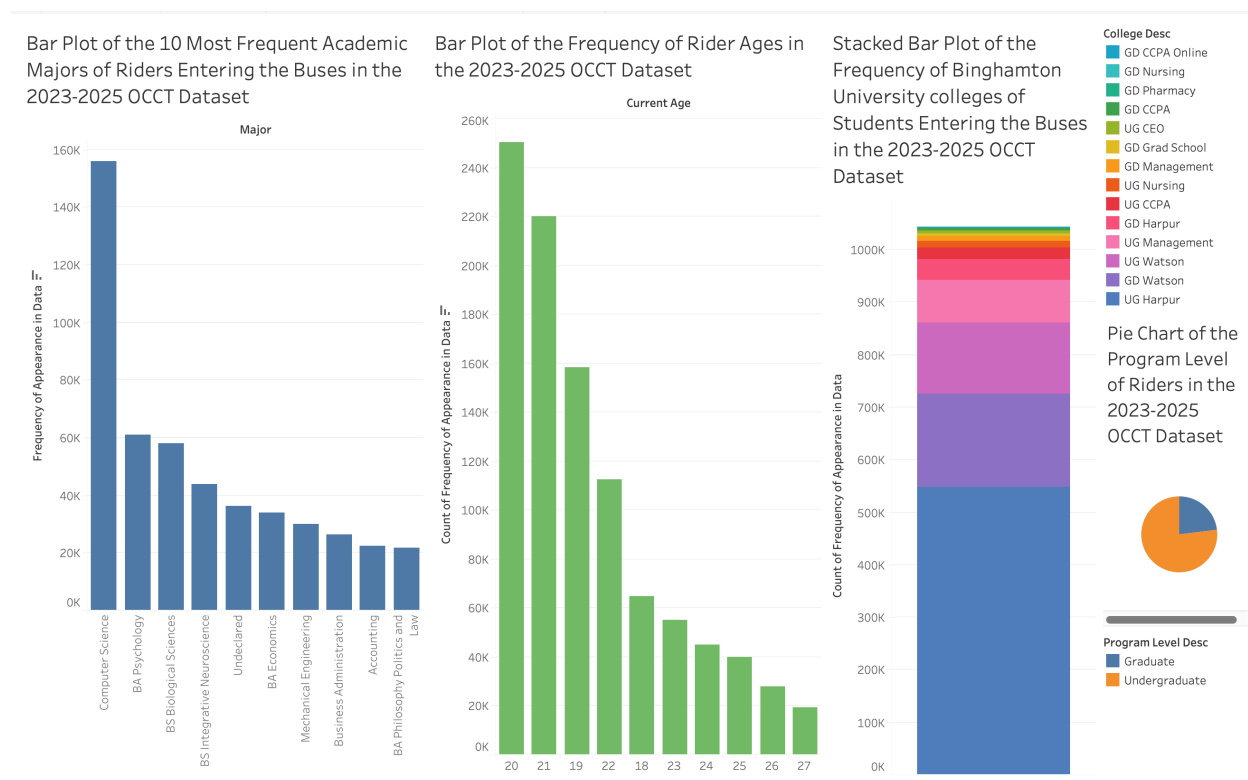


Figure 2: Dashboard of Rider Demographics in the 2023-2025 OCCT Dataset - Rider Demographics Interactive Dashboard

Figure 3 summarizes OCCT ridership patterns across time, including hour of day, day of week, season, and year. The heatmap shows that bus activity is highest during the regular weekday daytime and evening hours, especially from late morning through early evening. Ridership is much lighter overnight and on weekends, although some weekend activity still appears, particularly later in the day and evening. This pattern suggests that OCCT usage is closely tied to students' academic schedules, work schedules, and daily travel between campus and residential areas. The seasonal bar chart shows that Fall has the highest frequency of bus entries, followed by Spring and Winter, while Summer has much lower activity. This is expected because fewer students are in the Binghamton area during the summer term. The year pie chart also shows that most observations come from

2023 and 2024, with very little data from 2025, likely because the 2025 data was incomplete at the time of collection.

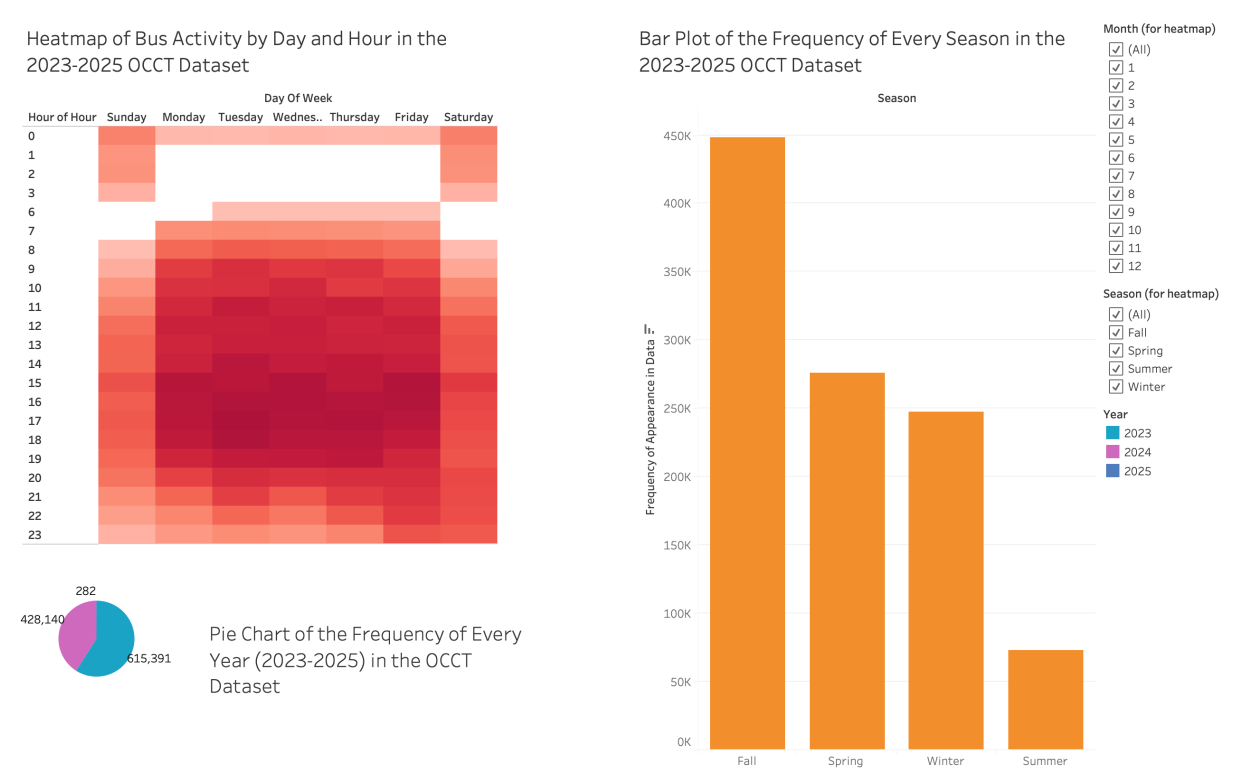


Figure 3: Dashboard of Season, Hour, Weekday, and Year Demographics in the 2023-2025 OCCT Dataset - Time Demographics Interactive Dashboard 1

Figure 4 shows how bus activity varies by both route and hour of day. Overall, the highest activity appears during the main daytime and evening periods rather than overnight. The Campus Shuttle shows consistently high activity across much of the day, especially from the late morning through the evening, reinforcing its role as one of the most frequently used routes in the system. The ITC–UCLUB route also shows strong activity during daytime hours, suggesting heavy travel between student residential areas and campus-related destinations. Westside Inbound appears busiest in the morning and early afternoon, while Westside Outbound becomes more active later in the day and evening, which is consistent with students traveling toward campus earlier and returning to residential areas later. Main Street, Downtown Center, and University Downtown Center routes show more moderate activity, with demand concentrated during specific parts of the day rather than evenly across all hours. Figure 4 specifically shows that route demand is not uniform: different corridors have distinct time-of-day patterns that likely reflect commuting, class schedules, and residential travel behavior. When coupled with Figure 3, this figure supports the idea that OCCT ridership is closely tied to student schedules.

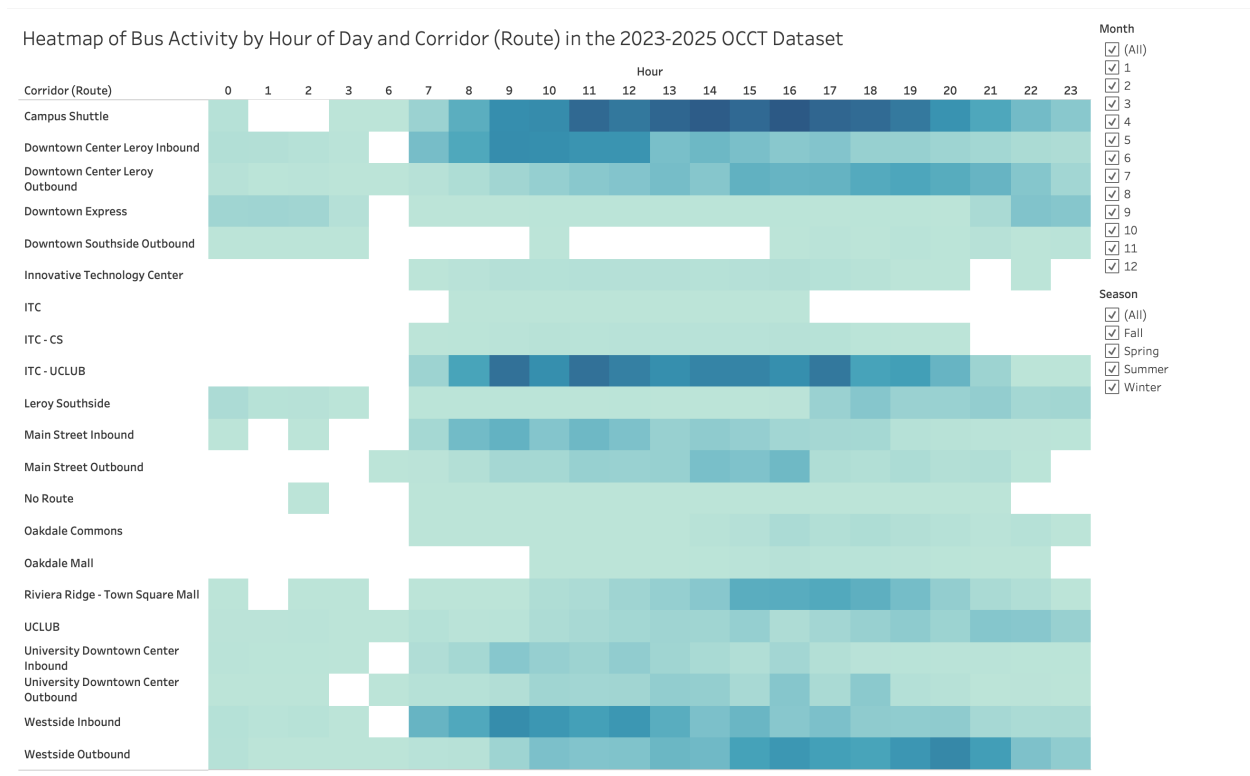


Figure 4: Dashboard of Hour and Corridor Demographics in the 2023-2025 OCCT Dataset - Time Demographics Interactive Dashboard 2

Figure 5 compares bus activity by hour for the ten most frequent stops, first including all stops and then excluding University Union. In the full stop heatmap, University Union clearly dominates the visualization, especially from late morning through the evening. Its activity is so much higher than the other stops that it makes the remaining stop-level patterns harder to see. This supports the earlier finding that University Union functions as the central hub of the OCCT system and may strongly influence stop-level analyses.

The second heatmap in Figure 5 removes University Union to make patterns among the other high-frequency stops more visible. Once University Union is excluded, stops such as Mohawk, UCLUB, Engineering Building, University Downtown Center, Hillside, and Rafuse show clearer variation by hour. Mohawk appears especially active from late morning through the afternoon, while UCLUB and University Downtown Center show stronger activity during the morning and early afternoon. This comparison shows why it is useful to examine stop-level activity both with and without University Union: including it captures the true dominance of the main campus hub, while excluding it reveals more detailed patterns among other important residential and academic stops.

Click the link below Figure 5 to see the complete heatmaps for all 24 hours.

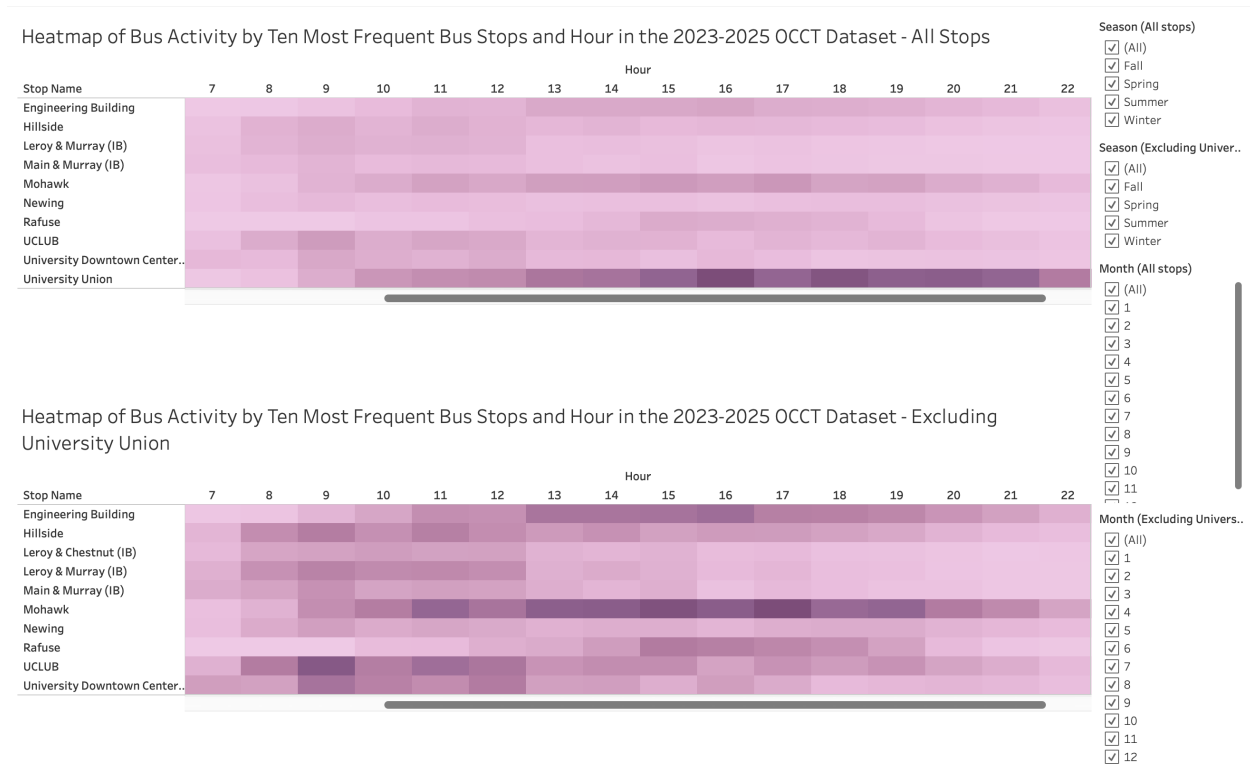


Figure 5: Dashboard of Hour and Corridor Demographics in the 2023-2025 OCCT Dataset - Time Demographics Interactive Dashboard 3

Figures 6, 7, 8, and 9 show the geographic distribution and density of OCCT boardings across the Binghamton area. In Figures 6 and 8, the University Union stop dominates the visualization, with the largest daily count. This confirms the earlier dashboard findings that University Union is the central hub of the OCCT system and that much of the ridership activity is concentrated around main campus. However, because University Union has such a high level of activity, it can make other important stops and neighborhoods appear less significant.

To address this, Figures 7 and 9 exclude the University Union stop, making the broader spatial ridership pattern easier to see. These figures show that activity is still highly concentrated around campus, but additional clusters become more visible in residential and student-heavy areas such as UCLUB/ITC, Hillside, Westside stops, Downtown Binghamton, and nearby campus stops like East Gym. The density maps reinforce this pattern by showing clear clusters of activity not only on campus, but also in off-campus residential corridors. Even without University Union, the densest part of the map remains in the center of campus, but also extends around student housing areas and downtown routes. This suggests that OCCT serves as an important connection between off-campus housing and the main campus.

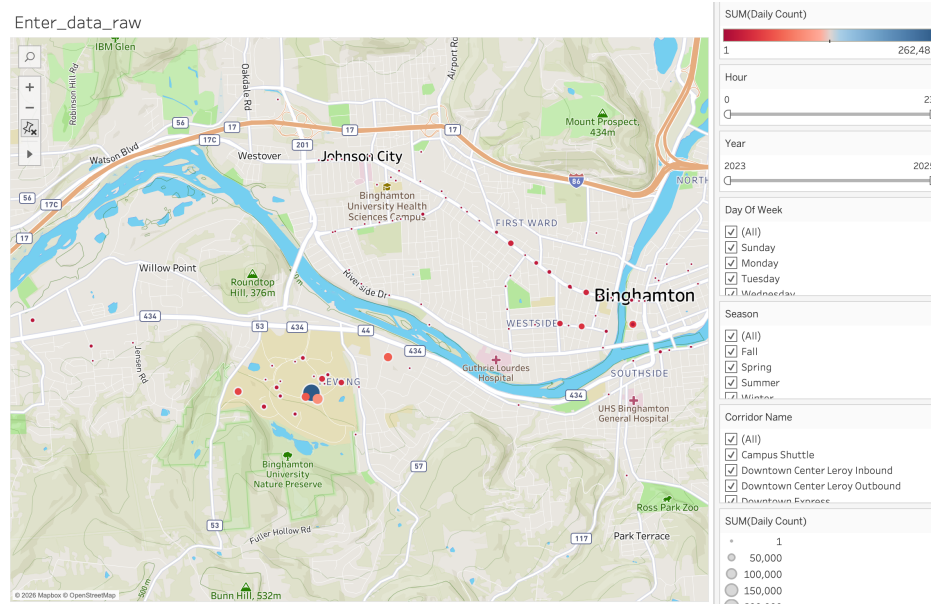


Figure 6: Dashboard of Frequency Count of Bus Entries in the 2023-2025 OCCT Dataset, with All Stops - Geographic Visualization Enter Count Data All Stops

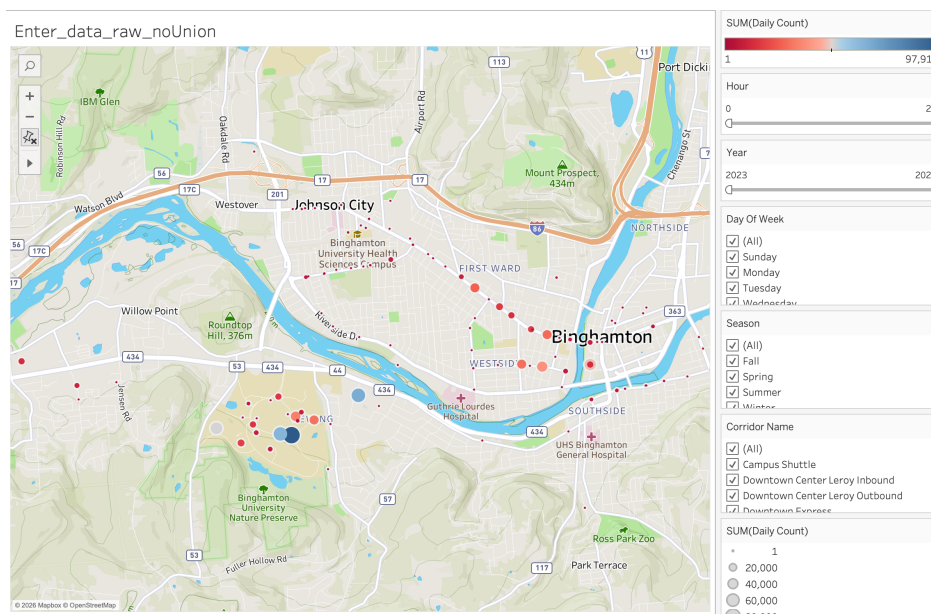


Figure 7: Dashboard of Frequency Count of Bus Entries in the 2023-2025 OCCT Dataset, without the University Union Stop - Geographic Visualization Enter Count Data without University Union

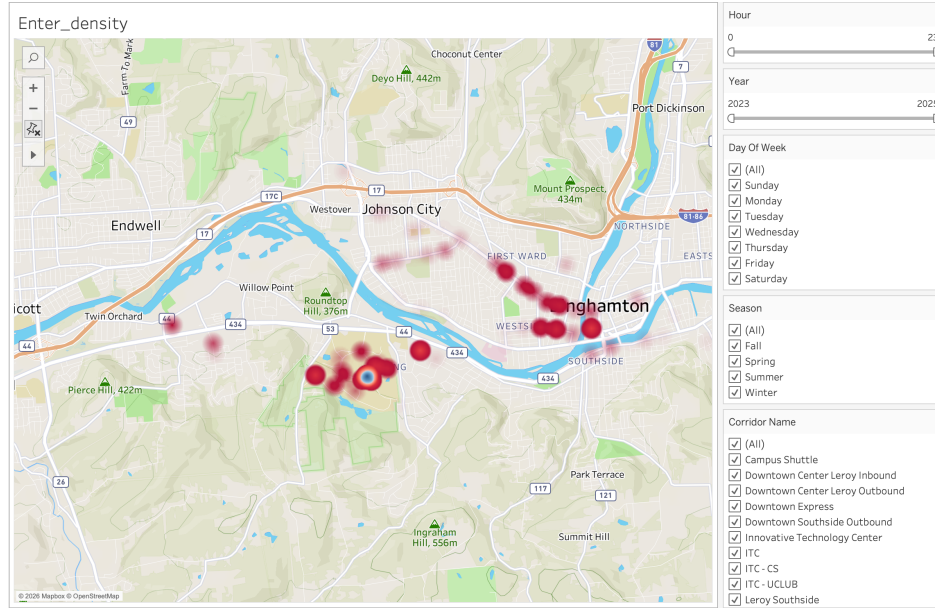


Figure 8: Dashboard of Density by Frequency Count of Bus Entries in the 2023-2025 OCCT Dataset, with All Stops - Geographic Visualization Enter Density Count Data with All Stops

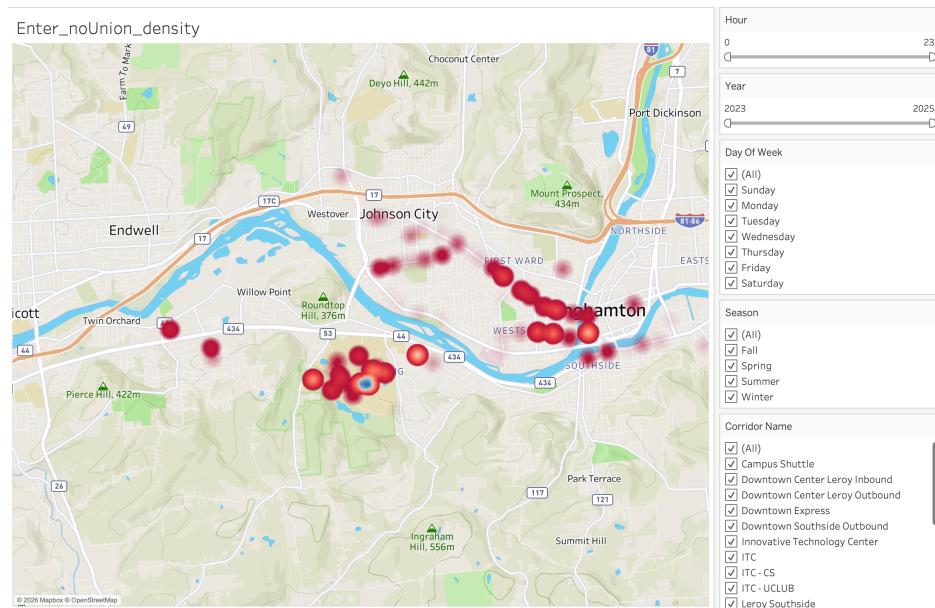


Figure 9: Dashboard of Density by Frequency Count of Bus Entries in the 2023-2025 OCCT Dataset, without University Union - Geographic Visualization Enter Density Count Data without University Union

Figures 10 and 11 show the geographic distribution and density of inferred exit locations across the OCCT system. Unlike the entry maps, which show where riders boarded, these visualizations estimate where riders may have gotten using on the exit-inference method presented in the previous section of this report. Thus, these

visualizations should be interpreted as approximate patterns rather than exact drop-off behavior. Overall, the exit maps show that inferred drop-offs are still heavily concentrated around central campus, especially near Mohawk and the Engineering Building. Unlike with the enter data, the University Union is not a major stop. However, Mohawk, Engineering Building, and University Union are all stops in central campus. This supports the idea that central campus, as a whole, is both a major boarding and destination location for OCCT riders.

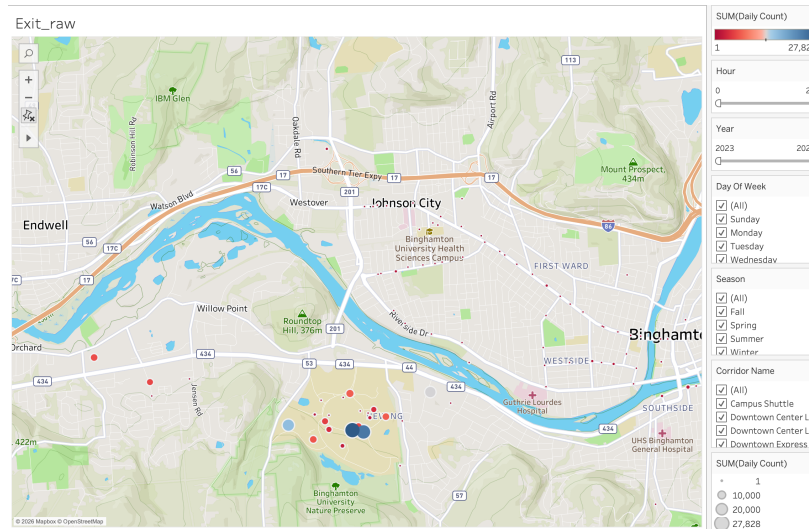


Figure 10: Dashboard of Frequency Count of Bus Exits in the 2023-2025 OCCT Dataset, with All Stops - Geographic Visualization Exit Count Data with All Stops

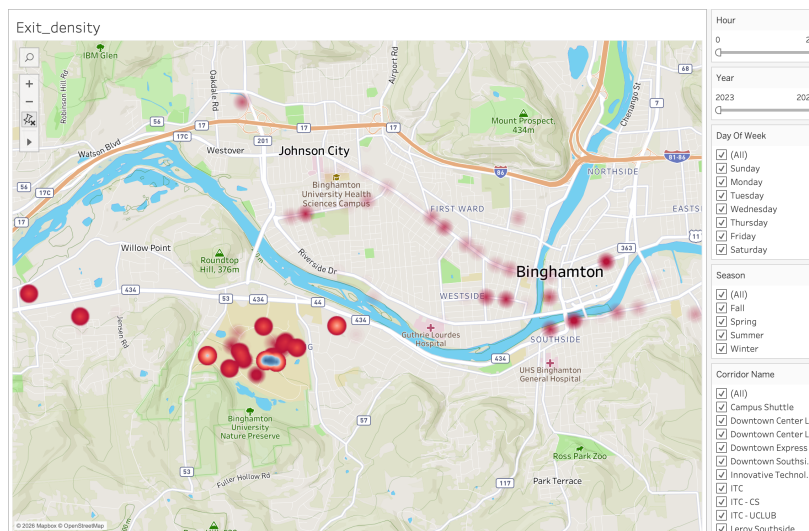


Figure 11: Dashboard of Density by Frequency Count of Bus Exits in the 2023-2025 OCCT Dataset, with All Stops - Geographic Visualization Exit Density Count Data with All Stops

Geo-spatial Analysis of a Proposed Bidirectional Shuttle Route

OCCT operates a Campus Shuttle to help students and other riders move around Binghamton University. Currently, the shuttle follows a continuous, one-directional loop. As a result, travel time can vary substantially depending on a rider's starting point and destination. The bidirectional Campus Shuttle analysis was conducted to determine whether the data shows evidence of demand for travel in the opposite direction of common Campus Shuttle trips.

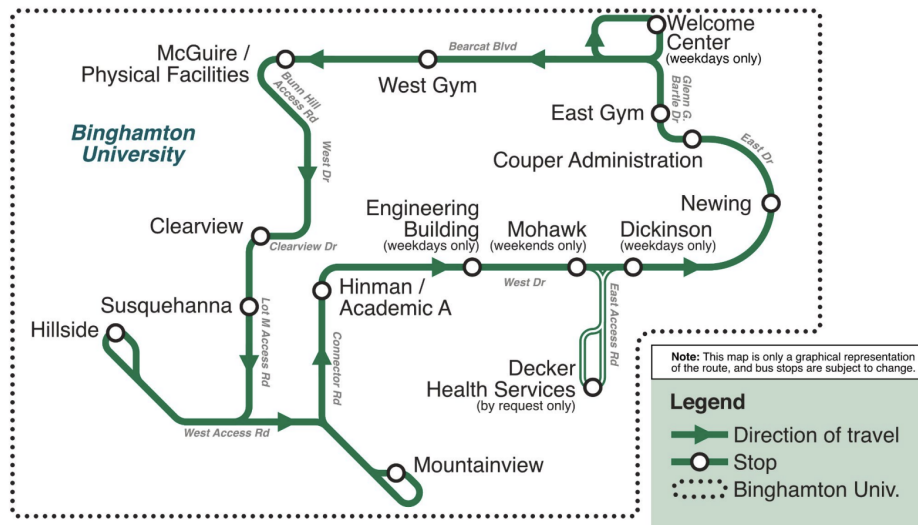


Figure 12: OCCT Campus Shuttle Route[1]

To determine whether the data provides evidence for bidirectional Campus Shuttle service, common inferred round trips taken by riders were identified. For example, a rider may travel from Susquehanna to the Engineering Building and later travel from the Engineering Building back to Susquehanna. If the route distance between these two stops is similar in both directions, then the current one-directional loop does not create a major travel imbalance for that pair. In that case, there is less evidence that bidirectional service would meaningfully improve the trip.

However, if the route distance is much longer from Susquehanna to the Engineering Building than from the Engineering Building to Susquehanna, or the opposite, then the current loop may create an inefficient return trip. In other words, riders may be able to travel from one stop to another relatively quickly, but the return trip may require them to remain on the shuttle for much longer because of the one-directional route design. These unequal stop-pair distances provide stronger evidence that bidirectional service could reduce travel time and improve rider convenience for certain Campus Shuttle trips.

In the original dataset, each record represents a bus entry, meaning the data shows where and when a rider boarded but does not directly record where that rider exited. Because of this limitation, exit locations had to be inferred using later rider activity. If a rider boarded at one stop and later, within the same day, entered the bus again at a

different stop, the later boarding location was used as an approximate indicator of where the earlier trip may have ended. Additionally, where the rider first entered the bus is assigned as the destination for the later trip. While this method does not perfectly reconstruct every rider's full trip, it provides a useful way to estimate common origin-destination patterns within the Campus Shuttle system. In addition to exact reverse matches, a 200-meter geographic tolerance was used to identify approximate reverse trips. If two inferred round-trip pairings involved stops that were not exact name matches but were within 200 meters, or approximately 0.12 miles, of one another, they were treated as the same general travel pattern and their counts were combined.

Figures 13 and 14 display the nine most common round-trips in the dataset. In figures 13, each row compares an initial trip, labeled as Origin 1 to Destination 1, with a corresponding reverse trip, labeled as Origin 2 to Destination 2. Both figures also include the real-world distance for each trip, with respect to roads, in miles and meters, along with an imbalance classification. The imbalance level was used to describe whether the forward and reverse patterns appeared similar enough to suggest balanced trips or whether one trip appeared much longer than the other. Figure 14 graphically shows how balanced each inferred round trip is.

Origin 1	Destination 1	Real Distance (miles)	Real Distance (meters)	Origin 2	Destination 2	Real Distance (miles)	Real Distance (meters)	Imbalance_Level
Hillside	Engineering	1.7	2735.88	Engineering	Hillside	2.55	4103.83	Moderate
Engineering	Hillside	2.55	4103.83	Hillside	Engineering	1.7	2735.88	Moderate
Hillside	Susquehanna	3.9	6276.44	Susquehanna	Hillside	0.35	563.27	Imbalanced
Hillside	Hinman	1.5	2414.02	Hinman	Hillside	2.75	4425.7	Imbalanced
Hillside	West Gym	3.1	4988.97	West Gym	Hillside	1.15	1850.75	Imbalanced
Hillside	Engineering	1.7	2735.88	Engineering	Susquehanna	2.2	3540.56	Moderate
Hillside	East Gym	2.6	4184.29	East Gym	Hillside	1.65	2655.42	Moderate
Susquehanna	Hillside	0.35	563.27	Hillside	Susquehanna	3.9	6276.44	Imbalanced
Hillside	Newing	2.1	3379.62	Newing	Hillside	2.15	3460.09	Balanced

Figure 13: Nine Most Common Inferred Round-Trips in the 2023-2025 OCCT Dataset

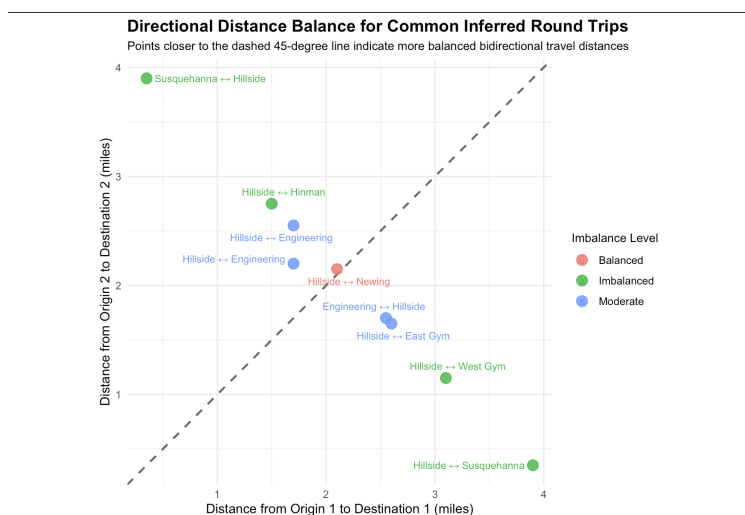


Figure 14: Directional Distance Balance for Common Inferred Round Trips

In the Figures 13 and 14, some Campus Shuttle stop pairs appear relatively balanced, while others show clear directional imbalance. For example, the Hillside to

Newing comparison is balanced, suggesting that, along the Campus Shuttle route, they are equidistant. In contrast, several Hillside-related pairs are labeled Imbalanced, including Hillside to Susquehanna, Hillside to Hinman, and Hillside to West Gym. These results suggest that, along the one-directional Campus Shuttle route, these stops are not equidistant from one another and riders traveling between the same two stops may experience substantially different travel distances.

The Hillside and Engineering comparison appears more moderate. Trips from Hillside to Engineering and Engineering to Hillside are both present, but the distances and classifications suggest that the relationship is not perfectly symmetrical. These pairs are labeled Moderate, meaning that while it may take longer to go from one stop to the other, the ride from the latter stop to the former on the one-directional Campus Shuttle route is not significantly shorter. Therefore, there may be some bidirectional demand, but the evidence is not as strong as it is for an imbalanced pair. This distinction is important because moderate imbalance does not necessarily mean a bidirectional shuttle would be ineffective; rather, it suggests that the service need may exist but may be more limited or dependent on time of day, class schedules, or specific student travel routines.

Overall, the results suggest that bidirectional demand exists within the Campus Shuttle system, but it is not evenly distributed across all stops. This means that a bidirectional Campus Shuttle may be useful in specific cases, but the data does not support treating all Campus Shuttle stops as having equal bidirectional demand. Instead, the strongest case for bidirectional service would come from identifying the stop pairs with repeated imbalanced or moderate patterns and then examining whether those patterns occur at consistent times of day. This is clarified by Figures 4 and 5. Figure 4 shows that Campus Shuttle activity is consistently highest from late morning through the evening, especially around 10:00 AM to 8:00 PM, while Figure 5 shows that major stops such as Mohawk, Newing, Hillside and Engineering Building remain active across much of the day.

Figure 15 further supports this timing argument by showing hourly Campus Shuttle activity across major stops. Although activity levels vary by stop, several stops are active during the same late-morning through evening period, especially between 10:00 AM and 8:00 PM. This overlap suggests that demand is not isolated to a single stop or route segment. Instead, multiple stops are being used at the same time, strengthening the case that a targeted bidirectional Campus Shuttle would be most useful during the shared high-activity window.

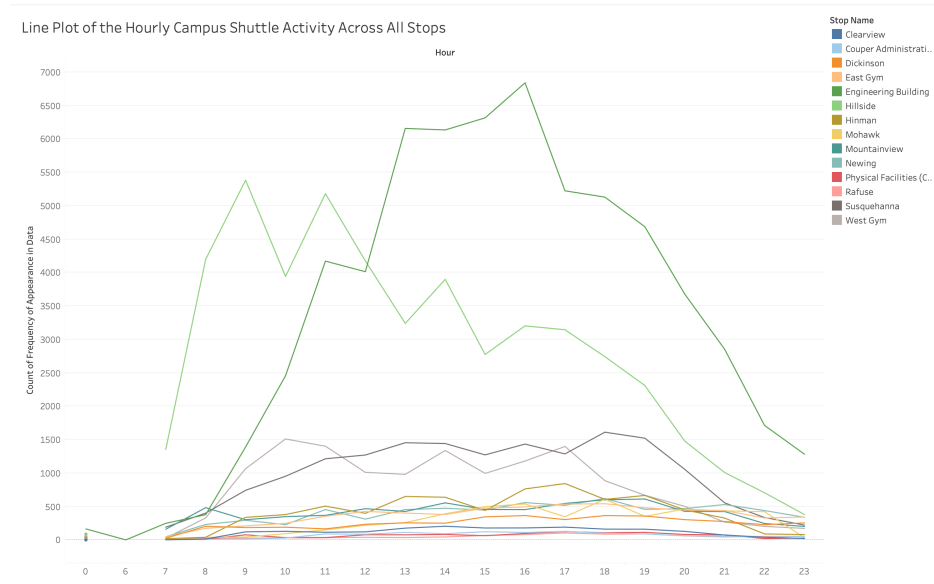


Figure 15: Line Plot of the Hourly Campus Shuttle Activity in the 2023-2025 OCCT Dataset - Hourly Campus Shuttle Activity

These findings should be interpreted with caution because the analysis depends on inferred exits rather than directly observed drop-off locations. A rider's next boarding location may not always represent the exact destination of their previous trip. For example, a student may walk more than 200 meters between locations before boarding again. Because of this, the bidirectional Campus Shuttle analysis should be viewed as an exploratory estimate of travel patterns rather than a definitive measurement of student movement. Still, the results provide a useful point for understanding where bidirectional service may be worth further investigation.

Statistical Evaluation of the Relationship Between the Main Street and Westside Corridors

This portion of the project examined the relationship between the Main Street and Westside corridors to determine whether ridership on one corridor changes in response to demand on the other. Specifically, this portion of the project evaluates whether Main Street ridership increases when Westside demand is lower, higher, or fuller.

It is important that the aforementioned question is answered because a simple comparison of total ridership is not enough to uncover the relationship between the Westside and Main Street corridors. Figure 16 and 17 demonstrates that overall, after controlling for the stop and hour, more riders in the dataset boarded the Westside than the Main Street corridor. Thus, Westside is more popular than Main Street in this dataset.

total_westside	total_main_st	mean_westside	mean_main_st	median_westside	median_main_st
46680	31209	5.36	3.58	4	2

Figure 16: The Popularity of the Westside and Main Street Inbound Corridors after Controlling Stop and Hour in the 2023-2025 OCCT Dataset

total_westside	total_main_st	mean_westside	mean_main_st	median_westside	median_main_st
40279	27404	14.9	10.1	10	6

Figure 17: The Popularity of the Westside and Main Street Outbound Corridors after Controlling Stop and Hour in the 2023-2025 OCCT Dataset

However, the more relevant planning question is not which corridor has more riders, but whether the two corridors respond to each other in ways that suggest substitution. If Main Street becomes more active when Westside is full or unusually busy, that could suggest that students use Main Street as an alternative when Westside service is strained. If Main Street becomes more active when Westside is less busy, that would suggest a different pattern: the corridors may serve related but distinct demand cycles rather than one corridor directly absorbing overflow from the other. Because both corridors serve major off-campus travel demand, understanding whether riders shift between them is important for interpreting congestion, route efficiency, and potential service adjustments.

To evaluate this, Main Street activity was compared against measures of Westside activity and Westside fullness. The analysis was conducted separately for inbound and outbound travel because the relationship between the corridors may differ depending on direction. For example, inbound service may reflect students traveling from off-campus housing areas toward campus, while outbound service may reflect students leaving campus at different times and for different reasons. Treating both directions as one combined pattern could hide meaningful differences in the way the corridors operate.

To investigate this relationship, a negative binomial regression model was fit to the data. The outcome variable was the number of Main Street Inbound/Outbound boardings observed for each shared stop during a given day-of-week-hour period. In the model, let Y_i represent the number of Main Street Inbound/Outbound boardings for observation i , where each observation corresponds to a shared stop, hour, and day-of-week combination. The model assumes:

$$Y_i \sim \text{Negative Binomial}(\mu_i, \theta) \quad (1)$$

where μ_i is the expected number of Main Street Inbound/Outbound boardings for observation i , and θ is the dispersion parameter. The negative binomial model allows:

$$\text{Var}(Y_i) = \mu_i + \frac{\mu_i^2}{\theta} \quad (2)$$

This means the variance can be larger than the mean. Because the response variable is a count, a standard linear regression model would not be appropriate: boardings cannot be negative, and ridership counts are usually right-skewed with many low-count observations

and a smaller number of high-count observations. A Poisson model is often used for count data, but it assumes that the conditional mean and variance are equal. In transit ridership data, this assumption is often too restrictive because ridership can vary substantially by stop, hour, and day of the week. Negative binomial models evaluate count outcomes and allow the variance to exceed the mean. Thus, a negative binomial model is optimal for this problem.

The expected number of Main Street Inbound/Outbound boardings was modeled using a log link function. The model estimated Main Street boardings as a function of Westside activity or fullness while controlling for major sources of routine variation. In particular, the model included controls for stop, hour, and day of week which helps separate the Westside-Main Street relationship from routine variation caused by location, time of day, and weekly travel patterns. All of this is reflected in Equations 3 and 4.

$$\log(\mu_i) = \beta_0 + \beta_1(\text{Westside Fullness}_i) + \beta_2(\text{Stop}_i) + \beta_3(\text{Hour}_i) + \beta_4(\text{DayOfWeek}_i) \quad (3)$$

Equivalently,

$$\mu_i = \exp(\beta_0 + \beta_1(\text{Westside Fullness}_i) + \beta_2(\text{Stop}_i) + \beta_3(\text{Hour}_i) + \beta_4(\text{DayOfWeek}_i)) \quad (4)$$

In this model, the key variable is Westside Inbound/Outbound fullness. Westside fullness was estimated for each boarding observation by tracking the running number of riders who had boarded a specific vehicle since its last arrival at the University Union. The running count for a given route loop was divided by the assumed bus capacity of 51 to create `westside.fullness.decimal`¹ This variable represents the estimated percentage of the bus that was full when a passenger boarded. The count was reset at the start of each new day to avoid carrying ridership totals across dates. In nearly all cases, estimated capacity stayed at or below 51 riders, with only four observations exceeding capacity.

The coefficient from the negative binomial model was interpreted using exponentiation. In other words, $\exp(\text{coefficient})$ gives the multiplicative change in expected Main Street boardings associated with a one-unit increase in the Westside fullness measure, holding stop, hour, and day of week constant. A value greater than 1 would suggest that Main Street boardings increase as Westside fullness increases, while a value less than 1 would suggest that Main Street boardings decrease as Westside fullness increases.

The inbound results provided good evidence of a possible relationship between the corridors.

westside_high	median_main_st	mean_main_st	median_westside_fullness	mean_westside_fullness	median_main_st_fullness	mean_main_st_fullness
FALSE	2	3.43	0.176	0.207	0.118	0.159
TRUE	2	3.78	0.275	0.3	0.118	0.166

Figure 18: Comparison of Main Street Ridership and Westside Inbound Fullness by Westside Inbound Demand Period

¹Off Campus College Transport states that blue buses seat 37 people and can stand 14 additional passengers for a total capacity of 51 passengers on trips within 10 miles. See <https://occtransport.org/pages/services/charter.html>.

	Estimate	exp(Estimate)	Std. Error	z value	Pr(> z)	Significance
(Intercept)	0.905106	2.47	0.057915	15.63	< 2e-16	***
westside_fullness_decimal	0.603127	1.83	0.05034	11.98	< 2e-16	***

Figure 19: Negative Binomial Model Results for the Relationship Between Westside Inbound Fullness and Main Street Ridership

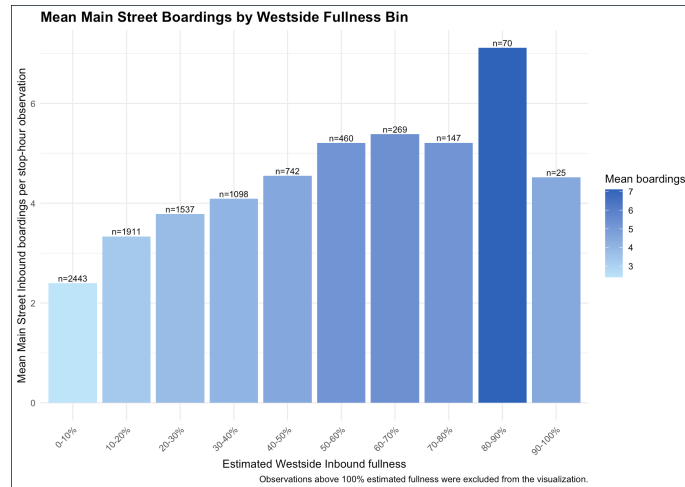


Figure 20: Mean Main Street Boardings by Westside Inbound Fullness, Binned by Current Westside Inbound Bus Fullness Capacity

Figure 18 suggests that Main Street Inbound ridership is slightly higher during high-Westside Inbound periods, but the difference is modest. When Westside Inbound was not classified as high, Main Street Inbound had a median of 2 boardings and a mean of 3.43 boardings per shared stop-hour observation. During high-Westside Inbound periods, the median remained 2, while the mean increased to 3.78. Westside Inbound fullness also increased from a median of 0.176 to 0.275, confirming that the high-Westside Inbound category captured periods of greater Westside Inbound demand. However, median Main Street Inbound fullness remained constant at 0.118 across both groups, suggesting that Main Street Inbound did not experience a large typical increase in crowding during high-Westside Inbound periods. Overall, the descriptive results point to a small increase in average Main Street Inbound activity when Westside Inbound is high, but not a strong shift in typical Main Street Inbound ridership.

However, as seen in Figures 19 and 20, the coefficient for Westside Inbound fullness was positive and statistically significant. After exponentiating the coefficient, the incidence rate ratio was 1.83, meaning that a one-unit increase in Westside Inbound fullness was associated with 1.83 times the expected number of Main Street Inbound boardings, holding stop, hour, and day of week constant. Since fullness is measured as a decimal, this corresponds to approximately a 6% increase in expected Main Street Inbound boardings for every 10 percentage-point increase in Westside Inbound fullness. This suggests that Main Street Inbound ridership tends to increase as Westside Inbound buses become fuller, providing evidence of a relationship between the two corridors.

This suggests that Main Street Inbound service may partially function as an alternative or complementary corridor when Westside Inbound demand changes. However, the interpretation should be cautious: this does not prove that individual riders are switching from Westside Inbound to Main Street Inbound. Instead, it shows an association between Westside Inbound conditions and Main Street Inbound boardings at similar times and under comparable stop/day/hour conditions.

westside_high	median_main_st	mean_main_st	median_westside_fullness	mean_westside_fullness	median_main_st_fullness	mean_main_st_fullness
FALSE	7	10.2	0.255	0.281	0.176	0.25
TRUE	4	10	0.373	0.394	0.216	0.283

Figure 21: Comparison of Main Street Ridership and Westside Outbound Fullness by Westside Outbound Demand Period

	Estimate	exp(Estimate)	Std. Error	z value	Pr(> z)	Significance
(Intercept)	-0.493325	0.61	0.438813	-1.124	0.260917	
westside_fullness_decimal	1.361546	3.91	0.084515	16.11	< 2e-16	***

Figure 22: Negative Binomial Model Results for the Relationship Between Westside Outbound Fullness and Main Street Ridership

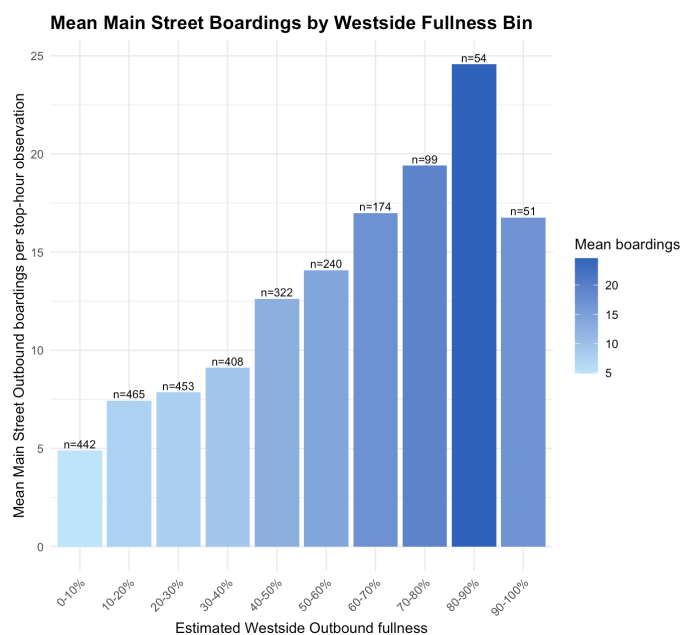


Figure 23: Mean Main Street Boardings by Westside Outbound Fullness, Binned by Current Westside Outbound Bus Fullness Capacity

The outbound results show a stronger model-based relationship between Westside fullness and Main Street ridership than the inbound results, but the descriptive summary is a little more nuanced. In Figure 21, when westside_high = FALSE, Main Street Outbound

had a median of 7 boardings and a mean of 10.2 boardings per stop-hour observation. When `westside_high = TRUE`, the median Main Street Outbound boardings decreased to 4, while the mean stayed almost the same at 10.0. This means the “typical” Main Street Outbound observation was not higher during high-Westside Outbound periods, but the average remained high, likely because some observations had much larger boarding counts.

Westside Outbound fullness clearly increased between the two groups. Median Westside Outbound fullness rose from 0.255 to 0.373, and mean Westside Outbound fullness rose from 0.281 to 0.394. Main Street Outbound fullness also increased modestly, with median fullness rising from 0.176 to 0.216 and mean fullness rising from 0.250 to 0.283. So, unlike the inbound case where Main Street Outbound median fullness stayed flat, the outbound case shows some increase in Main Street Outbound fullness when Westside Outbound is high.

The negative binomial model gives the strongest evidence of an outbound relationship. After exponentiating the coefficient for `westside_fullness_decimal`, an incidence ratio of 3.91 is achieved. This means that, holding stop, hour, and day of week constant, a one-unit increase in Westside Outbound fullness is associated with 3.91 times the expected Main Street Outbound boardings. Since fullness is measured as a decimal, a more realistic interpretation is that a 10 percentage-point increase in Westside Outbound fullness is associated with about:

$$e^{(1.361546 \times 0.10)} \approx 1.15 \tag{5}$$

or roughly a 15% increase in expected Main Street Outbound boardings. The p-value shows that this result is statistically significant.

For inbound results, the exponentiated coefficient was about 1.83, while for outbound, the exponentiated coefficient is 3.91, meaning the modeled association is substantially larger. In 10-percentage-point terms, the inbound model suggested about a 6% increase, while the outbound model suggests about a 15% increase. The descriptive results also differed. Inbound showed a small increase in mean Main Street boardings when Westside was high, from 3.43 to 3.78, while the median stayed at 2. Outbound, by contrast, had much higher Main Street boardings overall, but the median actually dropped from 7 to 4 when Westside was high, while the mean remained about the same. This suggests that outbound ridership may be more uneven or skewed: some high-ridership observations drive the model and the mean, even though the typical observation does not increase.

Conclusion

This project used exploratory data visualization, inferred travel patterns, and statistical modeling to better understand how students use OCCT service across routes, stops, times of day, and directions of travel. Overall, the findings show that OCCT ridership is highly concentrated around a small number of major routes, central campus stops, and recurring daily travel periods. Figure ?? demonstrated that the Campus Shuttle is one of the most frequently used routes in the dataset, while University Union functions as the central hub of the system. Other high-activity locations, including Mohawk, Engineering Building, Hillside, and UCLUB, University Downtown Center also appear

consistently across the Tableau dashboards. These patterns suggest that OCCT is not used evenly across the entire system; rather, demand is concentrated around major residential, academic, and campus-connection points.

Figures ??- ?? helped identify the central profile of main OCCT riders: undergraduate students in Harpur College and graduate Computer Science majors between the ages of 19 and 21 that live both on campus and around the Westside, who use the buses consistent with their academic schedules.

Figures ??, ??, and ?? further show that ridership is closely tied to student schedules. Activity is highest during weekday daytime and evening hours, especially from late morning through early evening, while overnight and weekend ridership are lower. Route-level heatmaps show that different corridors have distinct time-of-day patterns. For example, Westside Inbound is more active earlier in the day, while Westside Outbound becomes more active in the afternoon and early evening, consistent with students traveling toward main campus and then returning to residential areas. These findings support the broader conclusion that OCCT ridership reflects student academic schedules, residential patterns, and repeated daily travel routines.

Figures 7- 11 also reinforce this conclusion. Entry maps show that University Union dominates boarding activity, but once University Union is removed, other important spatial clusters become visible, including campus-adjacent stops, off-campus housing areas, and downtown routes. The inferred exit maps show that campus remains a major destination, especially near Mohawk and the Engineering Building. Although inferred exits are approximate, the combined entry and exit maps suggest that OCCT serves as a major connector between off-campus residential areas and the main campus, rather than simply serving isolated point-to-point trips.

The bidirectional Campus Shuttle analysis provided a more targeted look at whether the current one-directional Campus Shuttle loop may create inefficiencies for certain riders. Because the original data records boardings rather than exits, this portion of the analysis relied on inferred origin-destination patterns based on later-in-the-day boardings by the same rider. The results suggest that some Campus Shuttle stop pairs show evidence of repeated bidirectional demand, while others are more imbalanced, as seen in Figure 13. This means that the case for a bidirectional Campus Shuttle is not equally strong across all stops. Instead, the strongest evidence comes from stop pairs with repeated balanced or moderate reverse patterns, especially when those stops are active during the same time periods. Figure 14 showed that multiple major stops are active from late morning through evening, especially around 10:00 AM to 8:00 PM, suggesting that a bidirectional shuttle may be most useful if targeted to specific high-demand windows rather than operated uniformly throughout the entire day.

The Main Street and Westside corridor analysis showed that the relationship between these two corridors is more complex than simple route popularity. Westside had higher overall ridership than Main Street, but the statistical models showed that Main Street ridership tends to increase as Westside fullness increases, even after controlling for stop, hour, and day of week. For inbound service, the relationship was positive but modest: a 10 percentage-point increase in Westside fullness was associated with approximately a 6% increase in expected Main Street boardings. For outbound service, the modeled relationship was stronger, with a 10 percentage-point increase in Westside fullness

associated with approximately a 15% increase in expected Main Street boardings. However, the descriptive results show that this relationship should be interpreted carefully, especially outbound, where the median Main Street boardings decreased during high-Westside periods even though the model found a strong positive association. This suggests that some high-ridership stop-hour observations may be driving the modeled relationship.

Taken together, these findings suggest that Main Street may function as a complementary corridor to Westside during periods of higher Westside demand, but it should not be described as a direct replacement for Westside service. The relationship varies by direction and appears to depend on time, stop, and ridership intensity. Similarly, the bidirectional Campus Shuttle analysis suggests that bidirectional service may be useful in specific contexts, but the evidence does not support a blanket conclusion that all Campus Shuttle stops require equal bidirectional. Both analyses point to the same broader planning principle: OCCT service decisions should be made at the level of route, direction, stop, and time of day rather than from aggregate ridership totals alone.

There are important limitations to these findings. The dataset records boardings, not true exits, so all destination and reverse-trip analyses depend on inference. A rider's next boarding location may not perfectly represent where their previous trip ended, and some riders may walk between locations before boarding again. In addition, fullness estimates were based on running boarding counts and an assumed bus capacity, rather than direct onboard passenger counts. Although individuals are supposed to scan their university IDs when they enter the bus, many do not. As a result, the dataset may not capture every trip taken and likely undercounts total ridership. This means that the analysis reflects recorded bus entries rather than the full universe of OCCT usage. If non-scanning behavior varies by route, stop, time of day, or crowding level, then some patterns may be overrepresented or underrepresented in the data. Therefore, the results should be understood as a conservative estimate of ridership activity and as a tool for identifying meaningful trends, rather than as a complete measurement of all student transportation behavior.

Despite these limitations, the analysis provides several useful insights for future OCCT planning. First, University Union and other major campus stops should continue to be treated as central system hubs. Second, Campus Shuttle demand is high across multiple stops during shared late-morning through evening periods, making this time window the strongest candidate for further investigation into bidirectional service. Third, Main Street and Westside should be evaluated jointly because Main Street ridership is associated with Westside fullness, especially in the outbound direction. Overall, the results suggest that OCCT could benefit from targeted, data-informed service adjustments that account for direction, time of day, stop-level demand, and corridor interactions.

Acknowledgments

The author would like to thank Dr. Xingye Qiao for his guidance, support, and mentorship throughout this project. The author is especially grateful for his encouragement during the exploratory stages of the project and for his advice in connecting the technical findings to practical questions about campus transportation and student mobility. The author would also like to thank Dr. Megan McKinney and the

Individualized Major Committee for their time, effort, and advice throughout her IMP journey. From the initial Individualized Major pitch to the completion of this final draft, their advice and feedback have been deeply appreciated.

References

- [1] Off Campus College Transportation, Inc. Campus Shuttle Map.
<https://occtransport.s3.us-east-1.amazonaws.com/Spring+2026/OCCT+Spring+2026+Full+Master+Schedule.pdf>. Accessed May 11, 2026.
- [2] Off Campus College Transportation, Inc. Charter Service. *OCC Transport*.
<https://occtransport.org/pages/services/charter.html> Accessed May 11, 2026.

Appendix

- 1.) Github with all code used in this project:
https://github.com/mckenzie125/Bus_System_Analysis
- 2.) Tableau Visualizations:
<https://public.tableau.com/app/profile/mckenzie.skrastins6692/vizzes>